

SOLUTION

1: Hospital Doctor Shift Assignment

Solution

Step 1: Variables

- Doctors = D1, D2, D3
- Shifts = Morning (M), Afternoon (A), Night (N)

Step 2: Constraints

- D1 cannot work Night.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor is assigned exactly one shift.

Step 3: Backtracking Search

Step	Assignment	Constraint Check	Result
1	D1 → Morning	Valid	✓
2	D2 → Afternoon	Before D3	✓
3	D3 → Night	Not Morning	✓

No constraints are violated.

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Result: All constraints are satisfied using Backtracking Search.

2: Robot Grid Navigation

Solution

Grid Representation

S	□	□	X	□
□	X	□	X	□

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□  X  □  □  □
□  □  X  X  □
□  □  □  □  G

```

- S = Start
- G = Goal
- X = Obstacle
- □ = Free Cell

Constraints

- Move Up, Down, Left, Right.
- Cost = 1 per move.
- Cannot move through obstacles.
- Use Manhattan Distance heuristic.

BFS Node Expansion

Step	Expanded Node	Queue
1	S	(0,1),(1,0)
2	(0,1)	(1,0),(0,2)
3	(1,0)	(0,2),(2,0)
4	(0,2)	(2,0),(1,2)
5	(2,0)	(1,2),(3,0)
6	(1,2)	(3,0),(2,2)
7	(3,0)	(2,2),(4,0),(3,1)
8	(2,2)	Continue
9	Goal Found	Stop

Optimal Path

```

S
↓
↓
→
→
↓
→
↓
G

```

Total Cost

- Number of moves = **8**
- Total Cost = **8**

Result: BFS successfully finds the shortest path while avoiding obstacles.

Problem 3: Autonomous Rescue Robot

Solution

a) Constraint Analysis

Constraint	Effect on Decision
Move Up, Down, Left, Right	Limits movement options.
Blocked Cells	Robot must avoid obstacles.
Limited Sensing	Robot knows only nearby cells.
Movement Cost = 1	Encourages shorter paths.
Risk Zone Cost = +2	Robot prefers safer routes.
Dynamic Knowledge	Robot continuously updates the map.
Minimize Total Cost	Chooses the least-cost path instead of only the shortest path.

b) Solution Approach (Uniform Cost Search)

- Step 1:** Start at S.
- Step 2:** Expand the node with the lowest cumulative cost.
- Step 3:** Ignore blocked cells.
- Step 4:** Add neighbouring cells to the priority queue.
- Step 5:** If a risky zone is entered, add an extra cost of 2.
- Step 6:** Update the robot's knowledge after observing adjacent cells.
- Step 7:** Continue until the Goal (G) is reached.

Example Cost Calculation

Path	Cost
Normal Move	1
Risk Zone Move	3 (1+2)

The robot always selects the path with the minimum accumulated cost.

c) AI Concepts

Intelligent Agent

The rescue robot senses its surroundings, processes information, and performs actions to rescue survivors efficiently.

Rationality

The robot chooses the action that minimizes total cost while maximizing safety.

Environment Type

Property	Type
Observability	Partially Observable
Agents	Single Agent
Determinism	Partially Deterministic
Dynamics	Dynamic
State Space	Discrete

d) Justification

Uniform Cost Search is the most suitable approach because it always expands the path with the lowest cumulative cost. Since risky zones increase the travel cost, UCS finds the safest and least-cost path rather than simply the shortest one. The robot can also update its knowledge as it explores the environment, making it well suited for disaster rescue scenarios with limited information.